CHAPTER 4- BEGINNINGS OF CLASSICAL PHYSICS

- There was the class of geocentric and heliocentric model.
- Greek, Indian and Muslim scholars > Heliocentric , Western continued Geocentric.
- When clash occurred, the religious scripture prevailed. (winner)

- What did maths and geocentric model not agree on?
  Geocentric model could not agree on observations of math. For instance, retrograde is a problem that based on the difference between planet's orbits view and real ones. Another example is movements of planets. Some planets should always close to the sun. However sometimes they appear directly opposite the sun. So planets move different speed in their orbits.

- What is Galilean Relativity?
  The laws of physics are the same in all inertial referance frames, systems. Einstein's relativity is based on this principle.

- Explain relativity using a moving ship and observer on the shore.
  On a moving ship, a cannon ball is dropped from a mast.
  If observed from the shore, the cannon ball will have two motions. A forward motion provided by the ship and a downward motion caused by gravity.
  If observed from the ship, the cannon ball will have one motion that is downward by gravity.

- Copernicus paid by church while doing science. Yes it is interesting.

- The result of Copernicus' insistence of scientific measurement as a source of truth may well have been the starting point of the coming Scientific Revolution.

- The greatest problem Copernicus faced was to get the math to work.

- Copernic- Ordering the planets was very big achievement in that time.

- Copernicus model : Earth had to have two movements, one around the sun, other is its axis.

- There was conflict between copernicus and church.

- Giordano Bruno: Copernicus right. Sun was just a star. Millions of planet var zeki yaşamlı. Church killed him. Truth does not change whatever blablabla.

- Tycho Brahe: Observed during 30 years. And çok accurate modern telescopes bile onaylıyomuş. 6 ay boyunca yıldızları gözlemlemiş sonra dünyanın stationary sabit olduğuna karar vermiş. Sonra yaşlandıkça topladıkları dataları analiz edemeyeceğini düşünüyor. Then he recruited Johannes Kepler genç adamı işe aldıktan sonra Kepler kendi eksik kısımlarını önündeki çok accurate gözlemlerle tamamlıyor. Sonra bildiğimiz kepler yasaları.

- Why is Tyco Brahe important?
  So accurate. he used fixed sabit star to calculate velocity of a body.sonra vazgeçiyor tabi. Yukarıda yazıyor işte ölçümleri çok önemli.

- Kepler's contribution to science?
  His work on 3 laws of planetary motion, advancement of heliocentric model and accurate measurement.

- Kepler Archimedes gibi divided the circle into mant triangles to find ratio çevre ve çap arasındaki.
  Ve Kepler presume varsayarak marsın güneşin etrafında döndüğünü, yine üçgenlere bölüyor yörüngeyi. Her bir üçgeni geçme hızını ölçüyor.

- Kepler discovered that Tycho's data were not tutarlı consistent with a circular orbit. Tutarlı with an elliptical orbit.

- Explain 3 laws of planetary motion.
  1-planets move in elliptical orbit.
  2-same amount of time, same amount of area. Sweep- alan taramak kullanılabilir burda
  3- the distance to center and the period of planets movement/motion tkare r3 olayi sanirim.

- Cosmic whirlpool. Eliptik hareket suggested some kind of force acted on the planets. Depends on distance from the sun .yakın olduğunda daha güçlü bir kuvvet.
  He postulated sun rotated creating a vortex, a cosmic whirlpool ve gezegen carry eden.
  Then he learned Earth magnetism. Eğer gezegenler magnetse, sun büyük magnet. And thus what held the universe together was the simultaneous attraction and repelling of the various magnets on each other. Evreni ayakta tutan şey bu itme çekmeler oluyor felan dedi.

- How did Kepler's understanding shift from elliptical planetary motion to a magnetic theory of the universe's cohesion(uyumu)?
  Kepler initially proposed a vortex theory to explain planetary motion but later embraced the idea that celestial bodies, including the Sun, might function as magnets, suggesting a magnetic force as what held the universe together factor.

- Galileo Galilee
  his inquisitive mind would lead him in directions that clashed with prevailing culture. conflicted aristotle theories and put forward alternative approaches.

  He observed the motion of pendulums, noticed that time of swing was independent of the height from pendulum started swing. Başladığı yerden bağımsız olduğunu fark etmiş.
  Pendulum clock 2 pi kök l bölü g . the introduction of pendulum. The first harmonic oscillator used in timekeeping.

- Why is pendulum clock was important?
  Pendulum clocks increased accuracy of clocks enormously. 15 dk dan 15 saniyeye kadar.

Yani burada galilenin sarkaç araştırmaları, yükseklikten bağımsız olması ile güzel bir icat ortaya çıkmış bu da zamanı çok daha iyi ölçmemizi sağlamış.

- Galileo Galilee
  Yine başka bir bağımsızlık var. Bu kez eğik düzlemde hareketin ağırlıktan bağımsız olduğunu bulmuş.
  The motion was independent of Weight – Inclined planes.

- What was Galileos work with inclined plane? he observed gravity and acceleration of balls.

- Determined that speed is independent of weight of ball.

- The speed of rolling balls was proportional ,orantılı, to the time they fell.
  The distance traveled was proportional to the square of the time.

- Galileo tried different ones. 41 degree, 90 degree? Stayed the same.

- Galileo built his own telescope which he made amazing series of discoveries for universe.
  This series keşifler; bu arada telescope magnify 30 times// 30 kat büyütüyor.
  -The moon had craters, mountains, valleys.
  -Jupiter had four Moons satellite orbiting it.
  -The Milky Way was composed of hundreds of stars.
  -Venus had phases just like the moon evreler phases.
  -There were dark spots on the sun.

- Which systems is Galileo referring to in" The Dialogue Concerning the Two Chief World Systems? IS Geocentric and Heliocentric.

- Galileos ideas were not appreciated by the Vatican. Church was sensivite to any critisizm.
  Galileo's conflict with the Catholic Church symbolizes the clash between science and religion. Condemned by the Inquisition in 1633, he was made to retract his theories, barred from publishing his books, and put under house arrest for life. Galileo defended the Church against potential disbelief, asserting that when observation and evidence contradicted authority—be it Aristotle, the Church, or the Scriptures—authority should yield.

- He tried symbolize the Scientific Revolution. New world view.
  if dont get math could not discourse with nature

- Why the invention of telescope, microscope and accurate devices for measuring tempreture, pressure, and time is important?
  These technologic developments and improvements helped TRANSITION TO EXPERIMENTAL SCIENCE.

CHAPTER 5 – EXPERIMENTAL SCIENCE – THE SCIENTIFIC REVOLUTION

- Scientific Revolution Era:
  - -Belief natural law, math was language of nature.
  - -Rejection of authority, scholasticism
  - -belief in a mechanistic cosmos
  - -İnternational communities of scientist
    - *The Royal Society of London (1662)
    - *the Academie des Sciences de Paris (1666)
- Information Revolution and dissemination of it with improvement of printing technology.
  İt became difficult to control dissemination of information. Bilginin yayılması zorlaştı.
- How did the dissemination of information become easy? With printing tech.

- What is the difference between science and technology?
  Science needs to know and looking for casues. But technology runs independently from science and doesnt need to know why is it happening.

  EXAMPLES:
- Technology seems to develop independently of science and run well ahead of our scientific understanding of the technological processes
  - 1 - Steam technology developed well ahead of any understanding of thermodynamics.
- At other times, technological innovation clearly lays the foundation for scientific discovery.
  -2 - Galileo's telescopes: Galileo had no theory of optics to explain, or understand, how his telescope worked.

- How does magnifying glass büyüteç nasıl çalışır basic idea? :

- Object lens work?

- Kinds of telescopes? Microscopes?

- What is the difference between reflecting and refracting telescope?
  Refracting : Magnifying happens in eyepiece
  Reflecting: Mirror bounces light to the eyepiece
  Reflecting telescope uses a concave mirror to redirect light
  Refracting telescope uses convex lenses to bend light to towards a focal point.

- How to make telescope?

- A mechanistic world- 17 page

- Differe

CHAPTER 6 – SIR ISAAC NEWTON

- Explain Newton's Differantial Calculus.
- Why is Scientific Nomenclature important? How is it related to science?
  Science and scientific nomenclature improves each other. Science produce facts, ideas represent facts, and words express the ideas.
  These are 2 examples : Newton and Lavoisier.

CHAPTER 7- THE ENLIGHTMENT

- Natural Law-purely descriptive. Describe How nature behaved, not how we ought to live.
- one goal in discovering Natural law was be able to predict the future. Then confirmed .
- Halley's comet returns 1759 1835 1910 1986
- Natural law was linked with a search for greater determinism in nature

- Encyclopedia
- The first encyclopedia was edited by Denis Diderot and Jean D'Alembert.
- Most of the famous writers and philosophers of the 18th century contributed in one way or another.
- Encyclopedia became a secular substitute for Scriptures. It is the book of knowledge that will show you how reason

- What do we need to learn about history? What changed the world in terms of access to information?

  Maybe we can reach ancient tables to reach some informations. Invention of theatre (contribution to society) , invertion of books and establishing of libraries and then invention of encyclopedia. Additionally, nowadays this thing is definitely the internet. These are the cases that changed the game. And they are important dissemination of information easily.

- For some, The Principia was the New Testament, and the Encyclopedia was the new Scripture

  18th Century philosophers envisioned that reason, as the new scientific spirit, could also improve politics, morals, manners, the arts, literary criticism, and even public speaking.

- The enlightenment brought the Great Chain of Being crashing down (Aristotles ideas) . The consequences for religious authority and for civil law and order were great. *A hallmark of the enlightenment was the rejection of authority.*

- Locke: All knowledge comes from experience, a posteriori, rather than a priori. Humans are not inherently wicked or evil. Human nature is fundamentally good.
- The constituent parts of the society are engaged in machine like harmony - checks and balances -- it promotes maximum enjoyment of life, liberty, and prosperity

- ADAM SMITH: study history of economy, price of goods and labor, theories of rent and interest. Economical Perspective.
- According to Smith, wealth was enhanced by the division of labor.
  İş bölümü yaptıkça zenginlik artıyor, işçiler daha verimli çalışıyor ve daha üretken olabiliyorlar.
- Division of labor increases the overall wealth of nations, and thereby raises the standard of living of capitalists and workers alike.
- Smith : the state should avoid interference in the economy. Ekonomiye çok el sürmemeli. The invisible hand çok güzelregulator, düzenleyici+economic laws. Kendi kendine dönüyor zaten.
- He applied his theory to creating a price system. Doğru fiyat nasıl uygulanmalı . Arz talep Supply and demand diyor buna. Newtonun etki tepkisi gibi doğal.

- Novus Ordo Seclorum- New Secular World. Founders of US, embrace scientific revolution , while promoting the United States Constitution and the Bill of Rights
  Adamlar bilime inanıyor. Vizyonları geniş. Establish new country.

- LIGHT AND DARK- EPISODE 1 (BBC)

- According to Euclid, the light travels in straight lines.
  When a big thing is looked from far distance, it looks so small. So the light must travel in straight lines.
- Then Galileo used two lenses and installed them in a tube and called them objective and focal lenses. Then, for example, able to see ships 2 hours before. It promotes *some benefits in military, economy.*

Then observed these: Perfectly heavenly bodies is not perfectly-Mountains in moon.
4 satellite revolve around jupiter so world not center of universe.dispute geocentric
Galileo – Siderus Nuncius book
He is the first person that observe the sky in night without just naked eyes.

- Microscopic World- Hooke's discovery was not known to humans before that time.
  Hooke's book – Micrographia.
  Hooke saw animals function and great views from tiny animals.
  saw plant rectangles tissues and he called them 'cell' hücre.
  saw rainbow colors and thought they are added to the pure white color.

- Newton checked white color is pure? He used 2 prism. It was known that rainbow can be seen by transmitting white light. Newton obtain orange color and he pass it throught second prism. Saw no other color. *So he concluded that white color is made up other colors.*

- Later, Prism with Telescope was used to discover that there are gaps in spectra of Sun. This was then used to identify stars elements by Seki (1852) .

  Every Element absorbs certain wavelength producing a unique pattern of spectra.
  Her Element belirli bir dalga boyunu emerek benzersiz bir spektrum modeli oluşturur.

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